12.22 A Distribution With No Mean

WORKSHEET

The question

STEP III 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If (U, V) is a random point whose density depends only on its distance from the origin, what can you say about the angle Θ that the line from the origin to the point makes with an axis?

STAGE 1 The joint density

Covers: the joint density and radial symmetry

Ask yourself X and Y are independent, so multiply their densities. What does the result depend on?

STAGE 2 Using the symmetry

Covers: the identity for $\mathrm{P}(U/V < k)$

Pause and try $\frac{U}{V} < k$ can happen with $V > 0$ or with $V < 0$. Write each as a region. What does the rotation $(u, v) \mapsto (-u, -v)$ do to them?

STAGE 3 An angle

Covers: the density of U/V

Ask yourself Draw the region $\{V > 0, U < kV\}$. It is a wedge at the origin: what is its angle? With a radially symmetric density, the probability of a wedge is its angle divided by 2π. Then differentiate.

What's really going on

No mean

The mean of the Cauchy distribution would be $\int_{-\infty}^{\infty} \frac{k}{\pi(1+k^2)}\,\mathrm{d}k$, but $\int_0^K \frac{k}{1+k^2}\,\mathrm{d}k = \frac{1}{2}\ln(1+K^2) \to \infty$: the integral does not converge, and the mean does not exist. (Symmetry might suggest the answer is 0, but the positive and negative parts are both infinite, and $\infty - \infty$ has no value.) The variance is infinite too.

Worse, the average of n independent Cauchy variables is again Cauchy, with exactly the same distribution. (One way to see this uses the characteristic function $\mathrm{E}(e^{itX}) = e^{-|t|}$.) So averaging does not help at all: the average of a million measurements is no more reliable than one. Figure 12.22.1 shows what happens: the running average of normal samples settles down, while that of Cauchy samples lurches about for ever, jolted by occasional huge values.

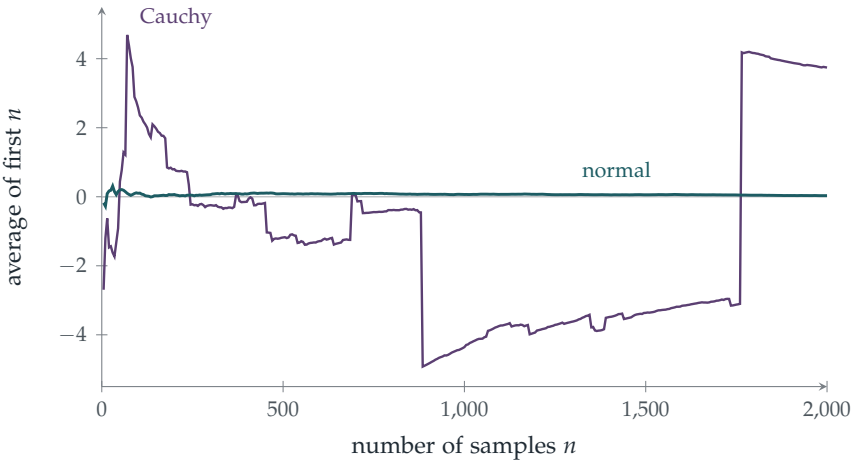


Figure 12.22.1 Running averages of 2000 independent samples. For standard normal samples (teal) the average settles down to 0, as the law of large numbers predicts. For Cauchy samples (plum) it never settles: every so often a single enormous value throws it somewhere new.

Where Cauchy variables come from

Shine a torch, pointing in a uniformly random direction, at a wall: the position of the spot on the wall is Cauchy distributed. In physics the same density, called the *Lorentzian*, describes the shape of spectral lines and resonance peaks (compare Section 11.1). In finance, heavy-tailed distributions like the Cauchy are used to model the occasional crashes that normal models wildly underestimate. The lesson for statistics is serious: methods based on averages and variances fail for heavy-tailed data, and robust methods, such as using the median (which for the Cauchy distribution is perfectly well behaved), are needed.

Poisson, Cauchy and a dispute

The distribution was studied by Siméon Denis Poisson in 1824, who noticed that it violates the law of large numbers. Augustin-Louis Cauchy met it in 1853 during a public dispute with Irénée-Jules Bienaymé over the method of least squares: Cauchy used it to show that least squares can be a poor method when errors are heavy-tailed. The name stuck to Cauchy, an example of what Stephen Stigler calls "Stigler's law of eponymy": no scientific discovery is named after its original discoverer.

Try this next

1 **The torch.** A torch at distance 1 from a long straight wall points in a direction making angle Θ with the perpendicular, where Θ is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Show that the spot's position $\tan\Theta$ has the Cauchy distribution.

2 **The reciprocal.** Show that if C is Cauchy, so is $\frac{1}{C}$. (*Hint:* $\frac{U}{V}$ and $\frac{V}{U}$.)

3 **The median works.** Show that the median of the Cauchy distribution is 0 and its quartiles are ± 1.

4 **Radius and angle.** For X, Y independent standard normals, show by changing to polar coordinates that $R^2 = X^2 + Y^2$ is exponential with mean 2 and is independent of the angle. (This is the basis of the Box–Muller method for generating normal random numbers.)

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on transformations of random variables.
- **Nassim Nicholas Taleb, *The Black Swan*** (Penguin). Heavy tails and why they matter, for a general reader.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press). Includes the Cauchy–Bienaymé dispute.

The full guided solution is in Section 12.22:
https://beyondstep.keithmcnulty.org/ch12/cauchy.html